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Computer Vision Project – Winter Term 2025/26

Face Recognition

Dr.-Ing. Thomas Köhler

Pattern Recognition Lab, Friedrich-Alexander-Universität (FAU) Erlangen-Nürnberg

e.solutions GmbH, Erlangen

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Face Recognition Example Use Cases

Unlock smart phones via face recognition¹



¹ <https://www.apple.com/iphone-x/#face-id>

Face Recognition Example Use Cases

Identification of occupants and personalization of vehicle settings



Scope of this Project

In the lecture:

- Learn how to design and evaluate the core of current facial recognition systems from a technical point of view
- Overview of modern machine learning methods in this field
- Discussion of their strengths and limitations

In the exercise:

- Implementation of a simple system comprising basic functionality for face verification, identification, and clustering
- Evaluation of face recognition algorithms
- Design own recognition method and participate in a challenge

Organization

- Exercises will be held virtually (start: CW 51/2025)
 - We will propose a couple of available time slots at the beginning of the week
 - Please drop me an e-mail with your availability
 - We will send you an invitation
 - **Be well prepared (only 15 - 30 minutes per group/week)!**
- Final submission via StudOn
 - Upload your source codes to StudOn **before the submission deadline!**
 - We will schedule a meeting to review your final solution
 - You will present your solution and we will ask questions
 - All exercise partners shall contribute equally to all tasks and shall be able to answer questions on each of the exercises!
 - **No certificate without presentation, review, and answering questions!**
- Additional exercise
 - Required to gain 10 ECTS
 - Will be a challenge with winners (details to be announced) :-)
 - Additional presentation and review

Outline

Introduction

Face Representation

- Eigenfaces

- Fisherfaces

- Deep Features

Selected Topics in Face Recognition

- Distance Measures and Face Verification

- Face Identification

- Face Clustering

- Evaluating Face Recognition Systems

Summary



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Introduction



The Face as a Biometric Marker

Biometric markers for identification²:

- Fingerprint
- Iris
- Speech
- Face

Face vs. other markers:

- Face the only marker that can be captured at large distances
- Can work with low-cost hardware

²Jain, Anil K., and Stan Z. Li. Handbook of face recognition. New York, Springer, 2011.

Challenges

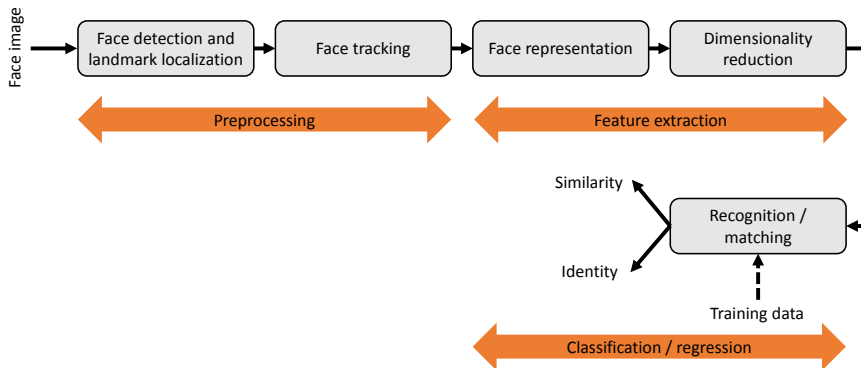
Intrinsic conditions:

- High intra-subject variation: expressions, changes in facial hair or pose, aging
- Low inter-subject variation: similar skin or hair color, similar eye glasses

Extrinsic conditions:

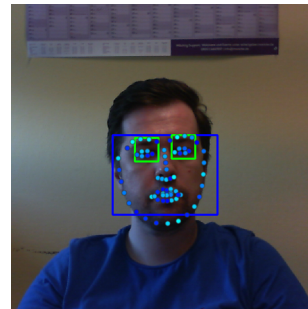
- Varying illumination: images captured at day and at night
- Image quality: facial data processed with different image or video codecs

Face Recognition Pipeline



Face Detection, Landmark Localization, and Tracking

- Detection of face region
- Detection of eye regions
- Extracting landmarks to model pose and facial expression
- Alignment using detected landmarks
 - Compensate for different head poses
 - Eyes, nose, and mouth at predefined positions
- Track bounding box and landmarks in video data



Face Representation

Geometric approach:

- Distances, areas, or angles between salient points (eyes, nose, mouth)
- Obtained by feature detection algorithms
- Requires hand-crafted features



Data-driven approach:

- Image intensities form raw features
- Learn suitable representation from exemplars
- Requires training from large datasets


$$\begin{pmatrix} 124 \\ 122 \\ 116 \\ 130 \\ 132 \\ \dots \end{pmatrix}$$

Face Recognition Problem Statements

Verification (one-to-one matching):

- Given one probe image and one gallery image
- Check if both images show the same identity

Identification (one-to-many matching):

- Given one probe image and a set of gallery images with identity labels
- Retrieve identity of probe image

Clustering (many-to-many matching):

- Given a set of unlabeled face images
- Cluster according to the identities captured in the data



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Face Representation



Data-Driven Face Representations

How to represent faces in digital images?

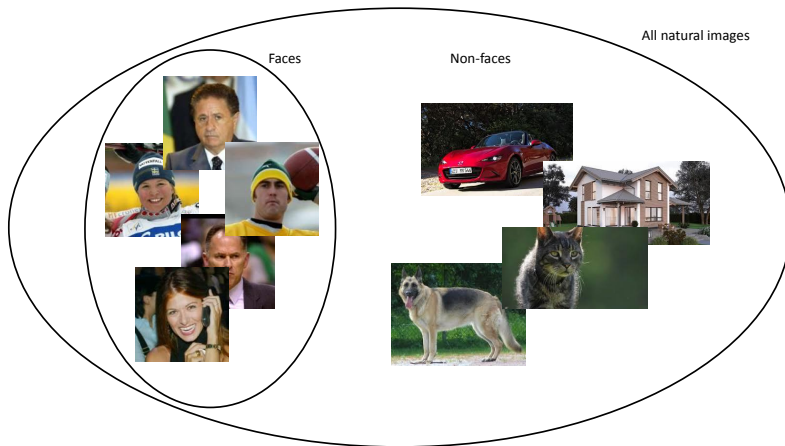


$$\begin{pmatrix} 124 \\ 122 \\ 116 \\ 130 \\ 132 \\ \dots \end{pmatrix}$$

- Grayscale image with $M \times N$ pixels and b bits per pixel (e. g. 64×64 , 8 bit)
→ $(2^b)^{M \cdot N}$ different images (e. g. $256^{4096} \gg 10^{1000}$)
- Only a small fraction of images corresponds to valid faces

Manifold of Face Images

Faces are a small subset of the natural image manifold



Learning Face Representations: Eigenfaces

- Let $\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}_i \in \mathbb{R}^D$ be a set of n face images with mean:

$$\mu_x = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \quad (1)$$

and covariance:

$$\Sigma_x = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \mu_x)(\mathbf{x}_i - \mu_x)^\top \quad (2)$$

- Project each $\mathbf{x}_i$ to a M -dimensional space ($M \ll D$):

$$\mathbf{y}_i = \mathbf{W} \mathbf{x}_i \in \mathbb{R}^M \quad (3)$$

- Seek linear transform $\mathbf{W} \in \mathbb{R}^{M \times D}$ that maximizes the variance of $\mathbf{y}_1, \dots, \mathbf{y}_n$:

$$\begin{aligned} \Sigma_y &= \frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i - \mu_y)(\mathbf{y}_i - \mu_y)^\top \\ &= \mathbf{W}^\top \Sigma_x \mathbf{W} \end{aligned} \quad (4)$$

→ Principal Component Analysis (Principal components $\equiv$ Eigenfaces)

Learning Face Representations: Fisherfaces

- Exploit two constraints (one unique face $\equiv$ one class):
 - Between-class scatter shall be maximum
 - Within-class scatter shall be minimum
- Seek the transform $\mathbf{W} \in \mathbb{R}^{M \times D}$ maximizing Fisher's linear discriminant:

$$J(\mathbf{W}) = \frac{\mathbf{W}^\top \boldsymbol{\Sigma}_{\text{inter}} \mathbf{W}}{\mathbf{W}^\top \boldsymbol{\Sigma}_{\text{intra}} \mathbf{W}} \quad (5)$$

Intra-class and inter-class scatter for c classes:

$$\boldsymbol{\Sigma}_{\text{intra}} = \sum_{i=1}^c \frac{1}{|\mathcal{X}_i|} \sum_{\mathbf{x}_k \in \mathcal{X}_i} (\mathbf{x}_k - \boldsymbol{\mu}_i)(\mathbf{x}_k - \boldsymbol{\mu}_i)^\top \quad (6)$$

$$\boldsymbol{\Sigma}_{\text{inter}} = \sum_{i=1}^c (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top \quad (7)$$

$\boldsymbol{\mu}_i$ is the mean of all faces $\mathcal{X}_i$ in the i -th class and $\boldsymbol{\mu}$ is the overall mean face

Learning Face Representations: Deep Features

Limitations of the methods discussed so far:

- Determine face representations under a linear model
- Limited robustness against pose or illumination variations

Extension:

- Use a non-linear face representation $f_{\theta}(\mathbf{x})$
- The transform $f_{\theta}(\mathbf{x})$ is implemented by a deep neural network
- Provides a face representation (aka. embedding)
- Parameters θ are learned from exemplars

Recap: Convolutional Neural Networks (CNNs)

Neural network:

- Computation graphs comprising neurons
- Propagation of input signals through the network to obtain outputs

Network design with input, output, and hidden layers:

- Convolutional layer: convolution of input neuron activation with filter kernel
- Pooling layer: fusion of clusters of input neuron activation
- Locally/fully connected layer: weighted sum of input neuron activation

θ : Parameters of all layers in the network

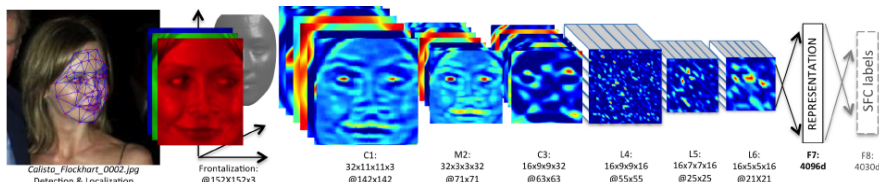
Face Representation via Classification

- Design neural network to classify face images according to their identity to one out of $c \geq 2$ classes
- Fully connected output layer $(\mathbf{W}, \mathbf{b})$ to model probability distribution:

$$p_i = \frac{\exp(\mathbf{w}_i^\top \mathbf{x} + b_i)}{\sum_{j=1}^c \exp(\mathbf{w}_j^\top \mathbf{x} + b_j)}, \quad i = 1, \dots, c \quad (8)$$

- Train network parameters on face exemplars by minimizing misclassification error (e.g. cross entropy loss)
- Activation of fully connected layer $\equiv$ face representation (aka. embedding)

Example: Facebooks DeepFace (trained on 4M faces)³



- Face frontalization for preprocessing
- Eight layer network: convolutional (C1, C3), max-pooling (M2), locally connected (L4, L4, L6), and fully connected layers (F7, F8)
- Feature maps describe face from high-level to low-level
- Activation of F7 fully connected layer used as face representation

³Taigman, Yaniv, et al. "Deepface: Closing the gap to human-level performance in face verification." Proceedings of the IEEE conference on Computer Vision and Pattern Recognition. 2014.

Face Representation via Contrastive Learning

Discussion of the classification-based approach:

- Classification requires class labels (identities)
- Models discriminative features for face recognition only implicitly
- Learned features are not necessarily optimal

Extension to regression-based / contrastive losses:

- Can be used for pre-training or fine-tuning
- No full labeling of dataset required

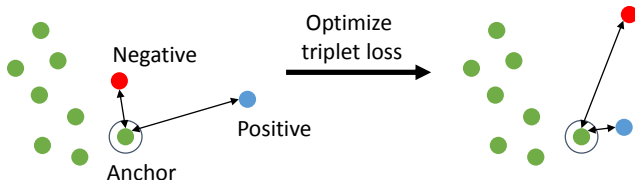
Triplet Loss

- Let $\mathbf{x}_i^a$ be an anchor face, $\mathbf{x}_i^p$ a face of the same subject (positive) and $\mathbf{x}_i^n$ be a face of different subject (negative)
- Ensure for all triplets $(\mathbf{x}_i^a, \mathbf{x}_i^p, \mathbf{x}_i^n)$ with margin α :

$$\underbrace{\|f_\theta(\mathbf{x}_i^a) - f_\theta(\mathbf{x}_i^p)\|_2^2}_{\text{anchor-to-positive distance}} + \alpha < \underbrace{\|f_\theta(\mathbf{x}_i^a) - f_\theta(\mathbf{x}_i^n)\|_2^2}_{\text{anchor-to-negative distance}} \quad (9)$$

- Penalize distances of positive and negative samples w.r.t. the anchor

$$\mathcal{L}_{\text{triplet}}(\theta) = \sum_{i=1}^n \|f_\theta(\mathbf{x}_i^a) - f_\theta(\mathbf{x}_i^p)\|_2^2 - \|f_\theta(\mathbf{x}_i^a) - f_\theta(\mathbf{x}_i^n)\|_2^2 + \alpha \quad (10)$$



Practical Considerations on Deep Features

Learning:

- Learn $f_{\theta}(\mathbf{x})$ on large datasets (Facebook: 4M faces, Google: 200M faces)
- Computationally very demanding
- Implemented on graphics processing units (GPU)

Inference:

- Determine face representation $f_{\theta}(\mathbf{x})$ by forward pass
- Efficient to compute using additional optimizations (weight quantization)
- Use face representation within lightweight classification or clustering models
- Can be implemented on embedded devices



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Selected Topics in Face Recognition



Face Verification

- Given two face images $\mathbf{x}_1$ and $\mathbf{x}_2$, is the image pair of the same person?
- Define a suitable distance measure $d(\mathbf{x}_1, \mathbf{x}_2)$, e.g., cosine distance:

$$d(\mathbf{x}_1, \mathbf{x}_2) = \frac{\|f_\theta(\mathbf{x}_1) - f_\theta(\mathbf{x}_2)\|_2^2}{\|f_\theta(\mathbf{x}_1)\|_2^2 + \|f_\theta(\mathbf{x}_2)\|_2^2} \quad (11)$$

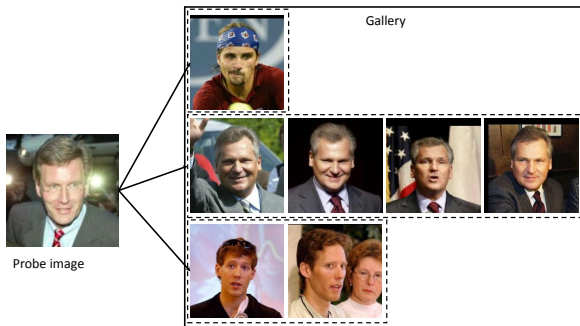
- Verification using the distance measure and threshold τ :

$$v(\mathbf{x}_1, \mathbf{x}_2) = \begin{cases} \text{same} & \text{if } d(\mathbf{x}_1, \mathbf{x}_2) \leq \tau \\ \text{not same} & \text{if } d(\mathbf{x}_1, \mathbf{x}_2) > \tau \end{cases} \quad (12)$$

- Alternatively, use similarity measure $s(\mathbf{x}_1, \mathbf{x}_2) = -d(\mathbf{x}_1, \mathbf{x}_2)$

Face Identification

- Given a probe image $\mathbf{x}_{\text{probe}}$ of unknown identity
- Determine identity $I(\mathbf{x}_{\text{probe}})$ from labeled images in a gallery



Two protocols:

- Closed-set: all possible identities of probes are contained in the gallery
- Open-set: identities of some probes are missing in the gallery

Face Identification with Closed-Set Protocol

Repeated face verification using k-nearest neighbors (k-NN) classifier:

- For the probe image $\mathbf{x}_{\text{probe}}$, find the k closest gallery images $\mathbf{x}_1, \dots, \mathbf{x}_k$ according to a distance $d(\mathbf{x}_{\text{probe}}, \cdot)$
- Identity of $\mathbf{x}_{\text{probe}}$ is the majority (mode) in $\mathbf{x}_1, \dots, \mathbf{x}_k$:

$$l(\mathbf{x}_{\text{probe}}) = \text{mode}(l(\mathbf{x}_1), \dots, l(\mathbf{x}_k)) \quad (13)$$

- Alternatively, we can consider similarities $s(\mathbf{x}_{\text{probe}}, \cdot) = -d(\mathbf{x}_{\text{probe}}, \cdot)$

Other discriminative classification models:

- Support vector machine (SVM)
- Random forests
- Boosting methods

Face Identification with Open-Set Protocol

Handle open space using thresholded nearest neighbors:

- Extract best matching gallery image $\mathbf{x}_{\text{match}}$ with minimum distance to $\mathbf{x}_{\text{probe}}$
- Possibly assign "unknown" identity to $\mathbf{x}_{\text{probe}}$:

$$I(\mathbf{x}_{\text{probe}}) = \begin{cases} \text{mode}(I(\mathbf{x}_1), \dots, I(\mathbf{x}_k)) & \text{if } d(\mathbf{x}_{\text{probe}}, \mathbf{x}_{\text{match}}) \leq \tau \\ \text{unknown} & \text{if } d(\mathbf{x}_{\text{probe}}, \mathbf{x}_{\text{match}}) > \tau \end{cases} \quad (14)$$

τ is the face verification threshold

Learning Open-Set Models using Known Unknowns

Extend the training setup of open-set models:

- Training set with KCs and known unknown classes (KUCs)

$$\mathcal{T} = \{(\mathbf{x}_i, y_i)\}_{i=1}^T \quad (15)$$

- Labels y_i can encode any KC ($\mathcal{C}_K = y_1, y_2, \dots, y_n$) or unknowns u

$$\mathcal{C} = \mathcal{C}_K \cup u \quad (16)$$

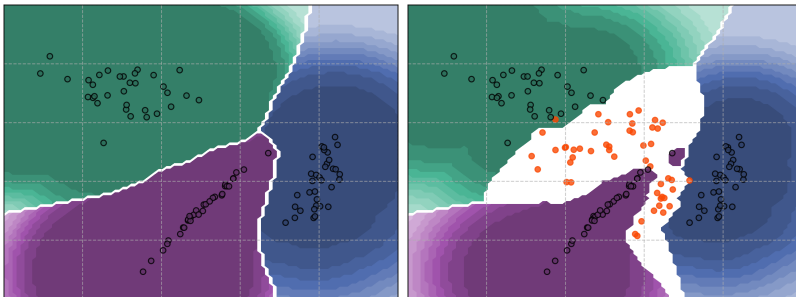
- Learn a likelihood function $h(\mathbf{x}; \mathcal{T})$ from the entire training set $\mathcal{T}$

Decision function to predict a label $\hat{y}$ with maximum likelihood:

$$\begin{cases} \hat{y} & \max_{y \in \mathcal{C}} h(\mathbf{x}; \mathcal{T}) \geq \tau \text{ and } \hat{y} \neq u \\ u & \text{otherwise} \end{cases} \quad (17)$$

Pseudo Labeling for Open-Set Learning

Idea: introduce pseudo labels for KUCs and learn from modified training set



Open-set decision boundary without KUCs (left) and with consideration of KUCs during training (right)⁴

⁴Koch, T., Riess, C., & Köhler, T. (2023). LORD: Leveraging Open-Set Recognition with Unknown Data. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 4386-4396).

Pseudo Labeling for Open-Set Learning

Single pseudo label (SPL):

- Assumption: all KUC samples form a large background class
- KUCs are treated as a single class with pseudo-label u resulting in the label set:

$$\mathcal{C} = \{\underbrace{y_1, y_2, \dots, y_n}_{n \text{ KCs}}, u\} \quad (18)$$

- Predicts unknowns directly independent of a decision threshold
- Can be used with any open-set / closed-set backbone to learn likelihood function

Pseudo Labeling for Open-Set Learning

Multi pseudo label (MPL):

- Assumption: every KUC sample models a different class
- KUCs are treated as separate classes with a single sample per pseudo label resulting in the label set:

$$\mathcal{C} = \underbrace{\{y_1, y_2, \dots, y_n\}}_{n \text{ KCs}}, \underbrace{\{y_{n+1}, y_{n+2}, \dots, y_{n+m}\}}_{m \text{ KUCs}} \quad (19)$$

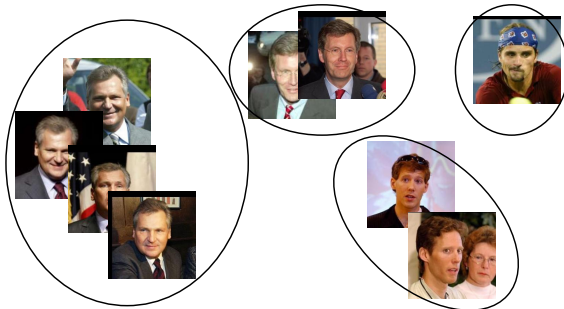
- Prediction of any pseudo label is mapped to the unknown class
- Can be used with any open-set / closed-set backbone to learn likelihood function but limited due to computational complexity

Face Clustering

- Given n face images $\mathbf{x}_1, \dots, \mathbf{x}_n$, cluster them into $k \leq n$ clusters $\mathcal{C}_1, \dots, \mathcal{C}_k$
- Minimize with cluster centers μ_i and distance measure $d(\cdot, \cdot)$:

$$(\mathcal{C}_1, \dots, \mathcal{C}_k) = \operatorname{argmin}_{\mathcal{C}_1, \dots, \mathcal{C}_k} \sum_{i=1}^k \sum_{\mathbf{x} \in \mathcal{C}_i} d(\mathbf{x}, \mu_i) \quad (20)$$

- k -means clustering: $d(\mathbf{x}, \mu_i) = \|\mathbf{x} - \mu_i\|_2^2$ and $\mu_i \equiv$ cluster mean



Clustering using k -Means Algorithm

Iterative algorithm:

1. Assignment: assign each face to the cluster with closest center

$$\mathcal{C}_i^t = \{ \mathbf{x} : \|\mathbf{x} - \mu_i\|_2^2 \leq \|\mathbf{x} - \mu_j\|_2^2 \text{ for all } i \neq j \} \quad (21)$$

2. Update: re-calculate cluster centers from current assignment $\mathcal{C}_1^t, \dots, \mathcal{C}_k^t$

$$\mu_i^{t+1} = \frac{1}{|\mathcal{C}_i^t|} \sum_{\mathbf{x} \in \mathcal{C}_i^t} \mathbf{x} \quad (22)$$

Evaluating Face Recognition Systems

Face identification with closed-set protocol:

- Rank of a probe image $\mathbf{x}_{\text{probe}}$ for gallery $\mathcal{G}$ with true matching image $\mathbf{x}_{\text{match}}$:

$$\text{Rank}(\mathbf{x}_{\text{probe}}) = \left| \left\{ \mathbf{x}' \in \mathcal{G} : s(\mathbf{x}', \mathbf{x}_{\text{probe}}) \geq s(\mathbf{x}_{\text{match}}, \mathbf{x}_{\text{probe}}) \right\} \right| \quad (23)$$

$\text{Rank}(\mathbf{x}_{\text{probe}}) = 1$ if $\mathbf{x}_{\text{probe}}$ is correctly associated with $\mathbf{x}_{\text{match}}$

- Rank- k Identification rate on a test set $\mathcal{T}$:

$$\text{IR}(r) = \frac{\left| \left\{ \mathbf{x}' \in \mathcal{T} : \text{Rank}(\mathbf{x}') \leq r \right\} \right|}{|\mathcal{T}|} \quad (24)$$

For $r = 1$ it is equivalent to the accuracy

Evaluating Face Recognition Systems

Face identification with open-set protocol:

- Trade-off between true identifications and unknowns that are incorrectly detected as knowns (false alarms) depending on similarity threshold θ
- Detection and identification rate for test set of knowns $\mathcal{K}$:

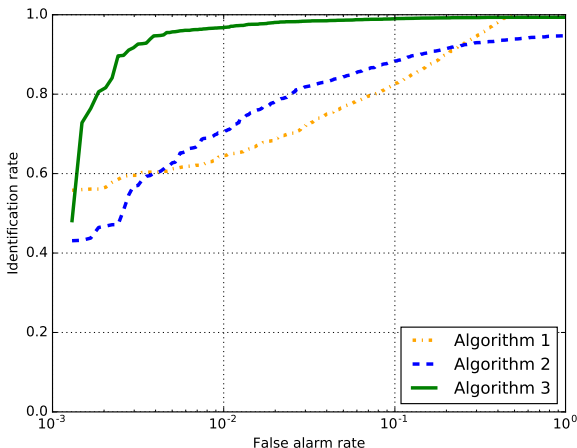
$$\text{DIR}(\theta) = \frac{|\{ \mathbf{x}' \in \mathcal{K} : s(\mathbf{x}', \mathbf{x}_{\text{gallery}}) \geq \theta \text{ and Rank}(\mathbf{x}') = 1 \}|}{|\mathcal{K}|} \quad (25)$$

- False alarm rate for complementary test set of unknowns $\mathcal{U}$:

$$\text{FAR}(\theta) = \frac{|\{ \mathbf{x}' \in \mathcal{U} : s(\mathbf{x}', \mathbf{x}_{\text{gallery}}) \geq \theta \text{ for any } \mathbf{x}_{\text{gallery}} \in \mathcal{G} \}|}{|\mathcal{U}|} \quad (26)$$

DIR Curve for Comparison of Face Recognition Algorithms

Depict $\text{DIR}(\theta)$ at different $\text{FAR}(\theta)$ with semi-logarithmic axes:





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Summary



Take Home Messages

- Widespread application domains of face recognition
- Still a hard problem under uncontrolled conditions (e. g. difficult poses)
- Face representation is a key component for modern face recognition systems
 - Learned on large training datasets
 - Different methodologies: Eigenfaces, Fisherfaces, deep features
- Recognition tasks (verification, identification, clustering) solved by common machine learning algorithms
 - Based on suitable face representation
 - Today also applicable on embedded devices

Further Readings

Overview on face image analysis and recognition techniques:

Anil K. Jain and Stan Z. Li. "Handbook of face recognition". Springer, 2011

Eigenfaces, Fisherfaces, and other classical methods:

Peter N. Belhumeur, João P. Hespanha, and David J. Kriegman. "Eigenfaces vs. Fisherfaces: Recognition using class specific linear projection." IEEE Transactions on Pattern Analysis and Machine Intelligence 19(7), 1997, 711-720.

Deep learning based methods:

- Yaniv Taigman *et al.* "Deepface: Closing the gap to human-level performance in face verification." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2014
- Florian Schroff, Dmitry Kalenichenko, and James Philbin. "Facenet: A unified embedding for face recognition and clustering." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015
- Yandong Wen *et al.* "A discriminative feature learning approach for deep face recognition." European Conference on Computer Vision, 2016.



Thanks for listening.
Any questions?